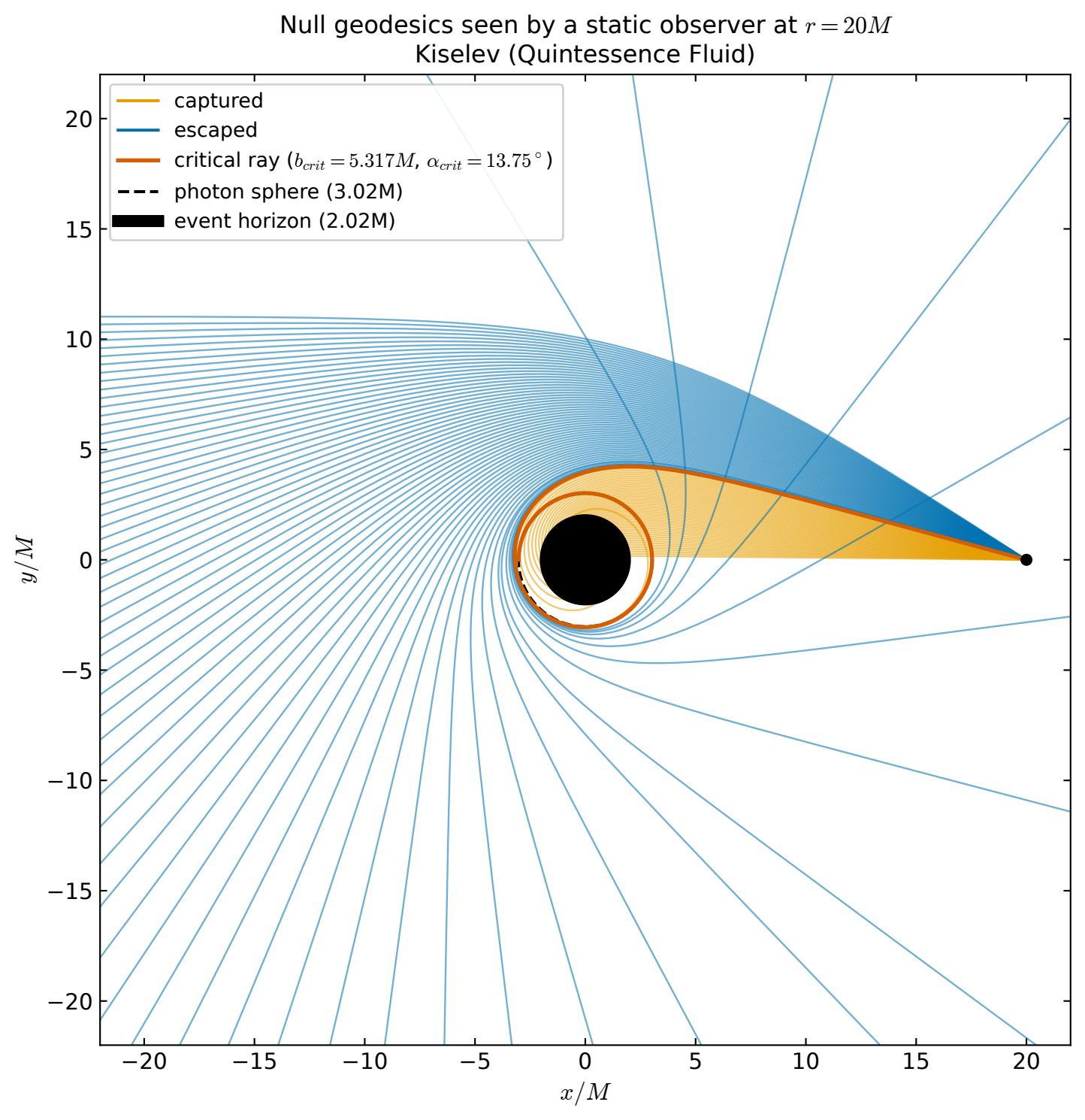

Null geodesics seen by a static observer at $r = 20M$
Kiselev (Quintessence Fluid)
captured
escaped
critical ray ($b_{crit} = 5.317M$, $\alpha_{crit} = 13.75^\circ$)
photon sphere (3.02M)
event horizon (2.02M)
y/M
x/M
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20